

on reaching the rural India. JNNSM is neutral to the choice of solar technologies and has provision for the development of all the viable technologies.

"Lot many initiatives like exclusive ministry with renewable energy focus and Jawaharlal Nehru Solar Mission are in execution in various phases and some states like Gujarat, Rajasthan and Haryana are taking the lead," adds Nayak.

There is a close synergy between semiconductor and solar PV technologies. To realise the shift to a decarbonised energy supply, solar PV is a key technology option for India. Ramakrishnan says, "The traditional sources of power generation, like thermal and hydro, have many structural issues resulting in project implementation delays. Therefore solar power projects are being fast-tracked by the government to bridge the demand-supply gap."

The current scale of solar PV manufacturing in India is small in comparison to the global standards, as majority of solar PV manufacturing in the country primarily comprises cell and module manufacturing with the bulk of value addition taking place outside the country.

"Although solar as an industry is still at a nascent stage in India compared to other markets, it is only going to get bigger," believes Singh.

Scope for a 'green' career

In a growing industry like solar power generation, opportunities will also naturally be on the rise.

"As per official statistics, in order to fulfil the 20,000MW installed capacity targets under the Jawaharlal Nehru National Solar Mission, the Indian solar energy industry will need an estimated 300,000 people by 2022 across all domains, profiles and levels," says Singh. Commenting on the scope for a career in this field, Nayak shares, "The scope is unlimited since PV offers a unique opportunity to solve the 21st century's energy and environment related problems simultaneously." He adds, "There are enough opportunities in production and project execution of

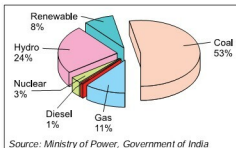


Fig. 1: Power generation capacity in India, by source of energy total 153 GW

solar projects as massive plans are being laid by every state."

The launch of JNNSM creates excellent opportunities for start-ups and entrepreneurs to jump into this industry, consequently creating jobs. "Once the government steps up the support and protection of the domestic solar manufacturing companies, we will see a rise in the manufacturing sector. If enough support is provided, in the near future, we might even face the shortage of skilled manpower," says Vineet Arya, head-marketing & communications, Tata Power Solar Systems.

With the recent launch of a Solar Research Energy Institute, and with more such research centres expected to be set up in India soon for both solar photovoltaic and solar thermal technologies, career opportunities in this sector are only going to increase.

"There will be demand for better electronics products like solar charge controllers, solar inverters and energy monitoring systems that are more suited for Indian conditions," says Ganesh Shankar. "There will also be a lot of scope for field work, which has more to do with connecting systems like solar panels, mounting structures, batteries, charge controllers, inverters and the load in setting up power plants," he adds.

Skills required

Skills required depend on the function to be played in the organisation. Usually, candidates are expected to be an expert in product development processes and troubleshooting machines, with a good insight into the manufacturing processes, analytical bend of mind and

also knowledge of financial aspects of project report preparation, cost estimation and proposal preparation. Knowledge of power electronics will be an added advantage as they will have to deal with power electronics systems like charge controllers and inverters.

"The skill set required in solar manufacturing units tends to be slightly different from the ones in other departments like marketing, sales and administration. For marketing and sales in the solar sector, an aspirant can focus on getting an MBA from a reputed college combined with the relevant business development experience in executing large projects," says Arya, adding, "For designing and developing core solar systems, a B.Tech/M.Tech graduate is desirable, who has the potential to design and develop systems. For installation, BE degree/diploma holders in civil infrastructure/switch yards are required, who can work as project managers."

The opportunities are unlimited as the potential of power generation is also unlimited. "A candidate would need diploma in engineering for standalone testing or integrated testing of a system, whereas a graduate in engineering with electrical and electronics is usually responsible for commissioning of large systems," says Nayak. He further states, "Postgraduates would be required to look at the higher value-adding possibilities, research and design of systems. PhD graduates would address optimisation of solutions and invention in bulk storage systems and new technologies like organic solar modules."

Most solar companies prefer a BE degree even for the position of a sales manager. They also expect the candidate to be enthusiastic and passionate about the industry, having a good understanding of the solar PV market, and capable of dealing with both domestic and international players.

Singh says, "A degree in engineering, finance, business development, quality control or marketing is the basic requirement. In addition, there is